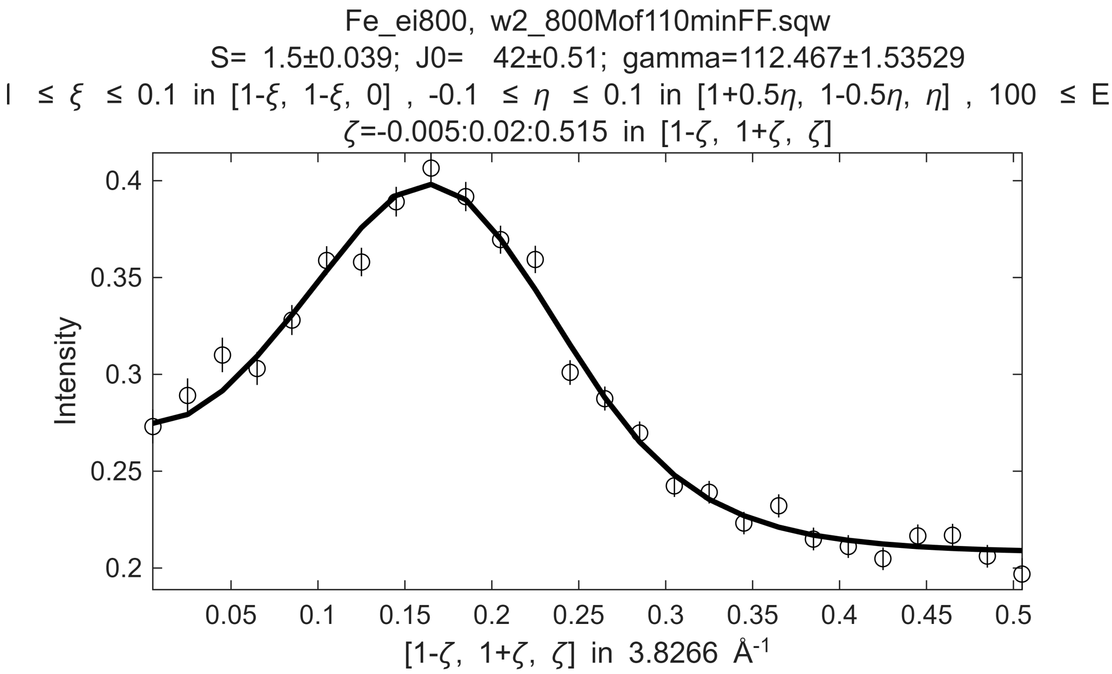
```
scuts_path = 'e:\SHARE\Fe\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
GHFFfitfile = fullfile(scuts_path,'e800GPffCuts.mat');
if ~exist('GHFFfitData','var')
    ld = load(GHFFfitfile);
    GHFFfitData = ld.GHFFfitData;
end
```

```
mi = maps_instrument(787,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);

cut_names = fieldnames(GHFFfitData);
for i=1:numel(cut_names)
    GHFFfitData.(cut_names{i}) = GHFFfitData.(cut_names{i}).set_instrument(mi);
    GHFFfitData.(cut_names{i}) = GHFFfitData.(cut_names{i}).set_sample(sample);
end
```

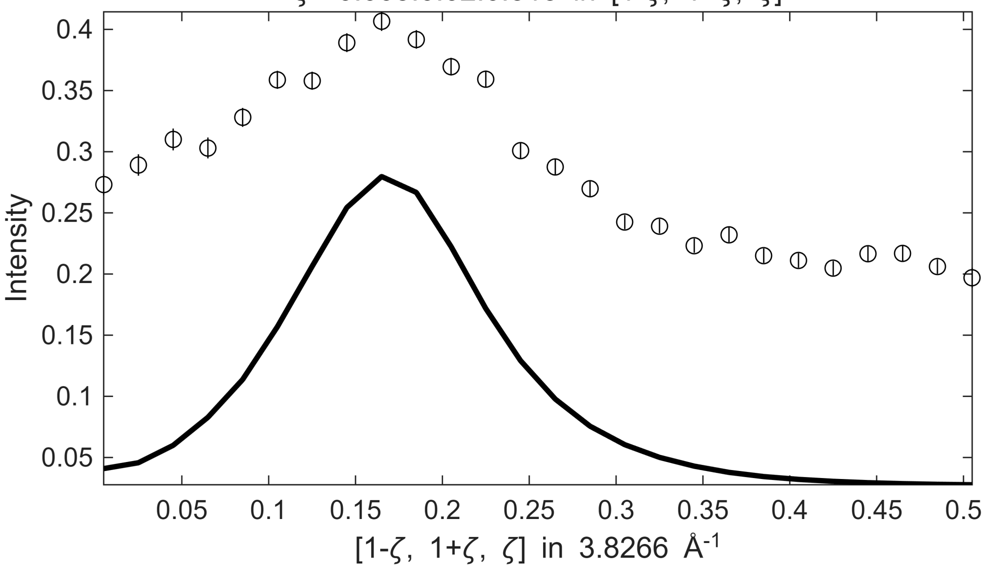
```
dir_name  = "GP";

dE_step = 4; %original energy transfer step data were binned to. No point in
going finer
half_dE = 5; % half width of data binning

if exist("res_name",'var') && ~exist('fit_reEi800GP','var')
    ld = load(res_name);
    fit_reEi800GP = ld.fit_reEi800GP;
else
    fit_reEi800GP = struct();
end


for i=1:numel(cut_names)
    acolor k
    fprintf('******************************\n')
    fprintf('**** CUT: %s\n',cut_names{i});
    fprintf('******************************\n')

    cut_to_work = GHFFfitData.(cut_names{i});
    if isfield(fit_reEi800GP,cut_names{i})
        continue;
    end
    all_en = cut_to_work.data.p{2};
    en_range = min_max(all_en);
    cut_en = 100+half_dE:10:en_range(2)-half_dE;
    [fr_800const_bg,res_name] = fit_multicuts_along_direction(...
        {cut_to_work},'FitEn_Ei800ff_1D',dir_name, ...
        1,cut_en,dE_step,half_dE,true);
    fit_reEi800GP.(cut_names{i}) = fr_800const_bg;
```

```
    acolor g
    bg_data = extract_bg_par(fr_800const_bg.all_fit_par,[1,2]);
    % original slope parameters

pd(bg_data(1));lx(fr_800const_bg.all_fit_par(1).en,fr_800const_bg.all_fit_par(end
).en);keep_figure; ly 0 3;

pd(bg_data(2));lx(fr_800const_bg.all_fit_par(1).en,fr_800const_bg.all_fit_par(end
).en);keep_figure

    save(res_name,'fit_reEi800GP');
end
```

```
******************************
**** CUT: Ei800_off110minFFGP
******************************
******************************
**** En = 105±5 going to 245
******************************
```

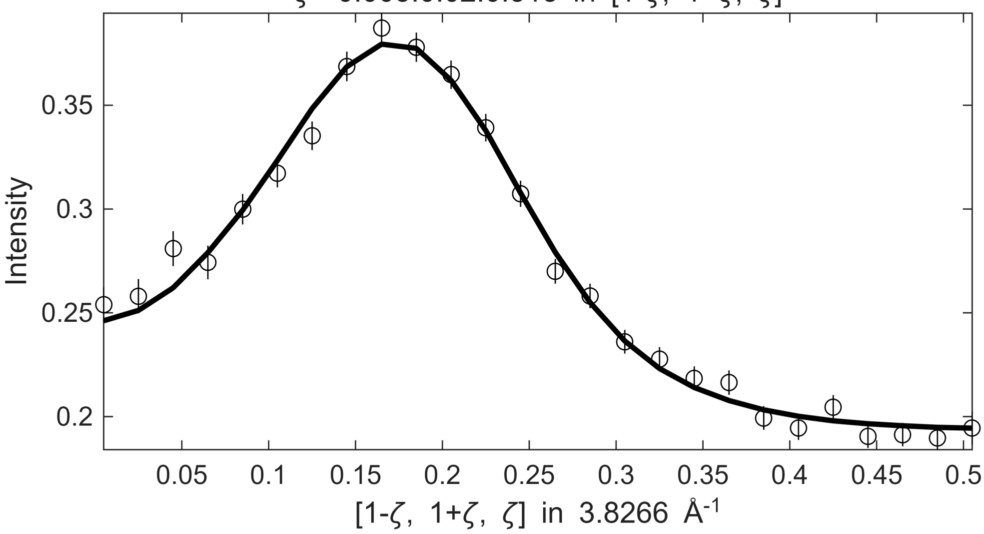


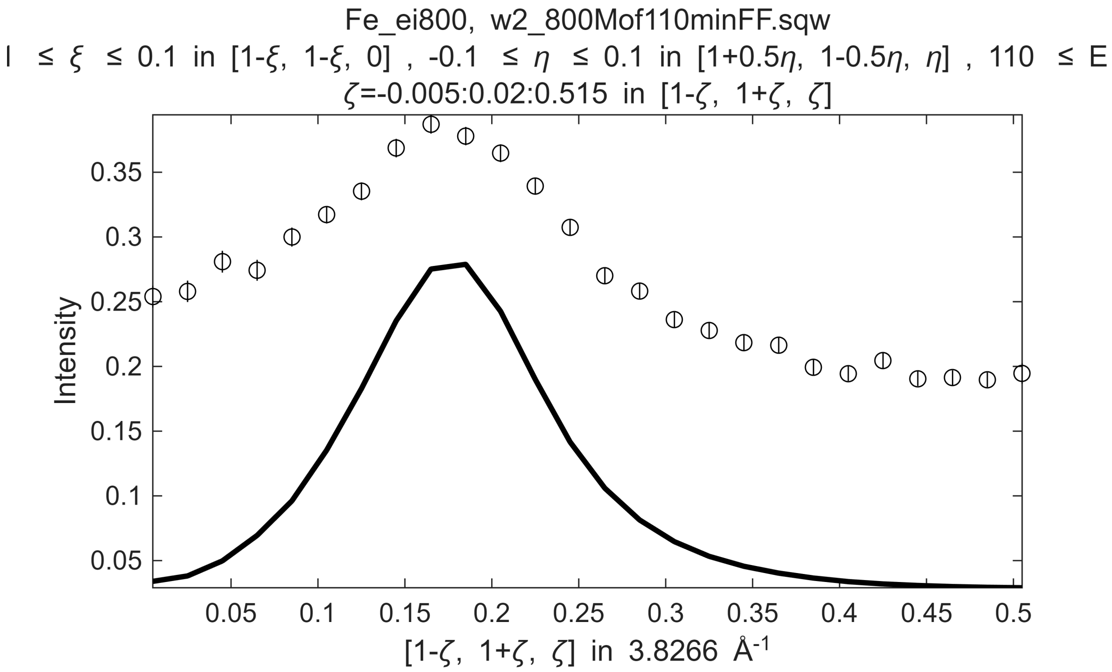
Fe_ei800, w2_800Mof110minFF.sqw
I $\leq \xi \leq$ 0.1 in [1-ξ, 1-ξ, 0] , -0.1 $\leq \eta \leq$ 0.1 in [1+0.5η, 1-0.5η, η] , 100 $\leq$ E
ζ=-0.005:0.02:0.515 in [1-ζ, 1+ζ, ζ]



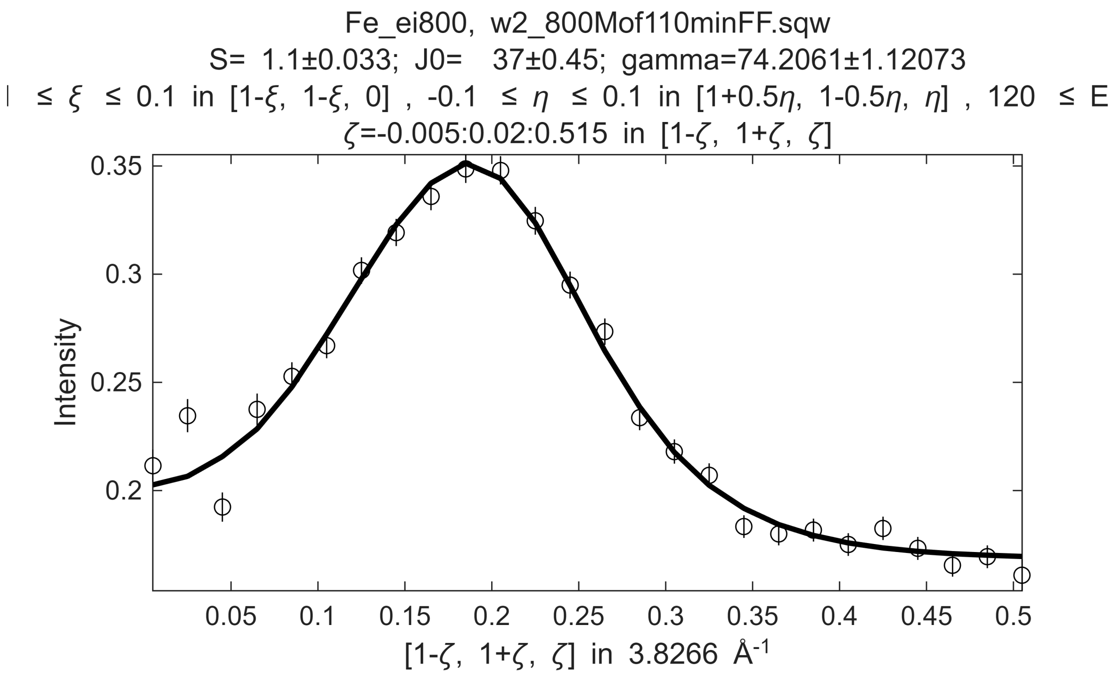

```
******************************
**** En = 115±5 going to 245
******************************
```

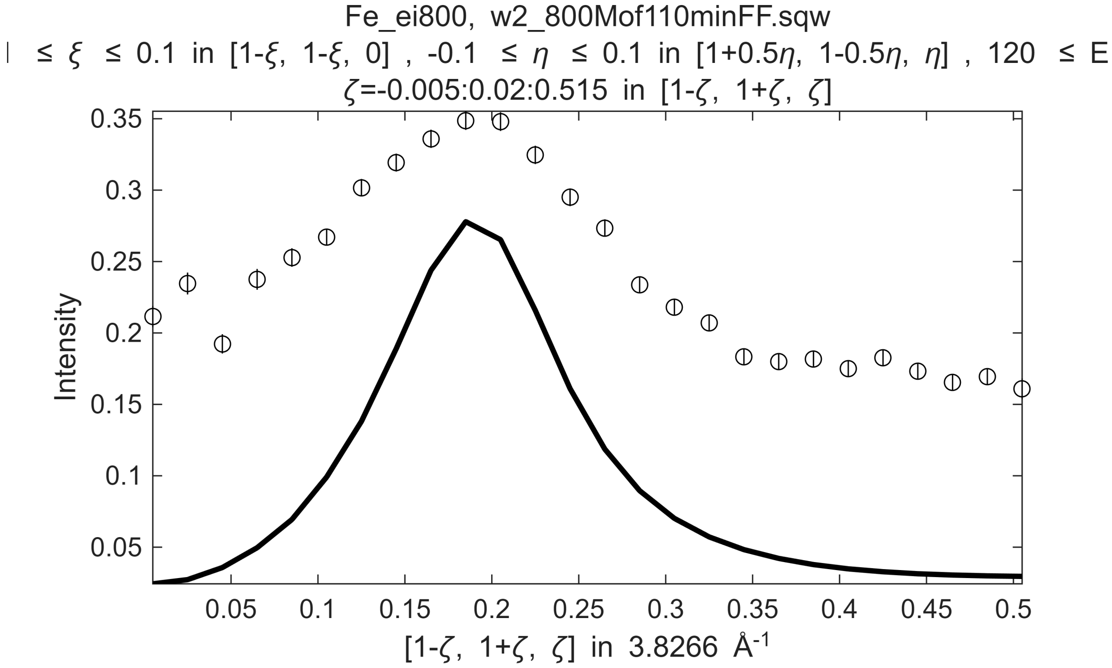
Fe_ei800, w2_800Mof110minFF.sqw
S= 1.4±0.028; J0=   40±0.35; gamma=94.9313±1.10934
I $\leq \xi \leq$ 0.1 in [1-ξ, 1-ξ, 0] , -0.1 $\leq \eta \leq$ 0.1 in [1+0.5η, 1-0.5η, η] , 110 $\leq$ E
ζ=-0.005:0.02:0.515 in [1-ζ, 1+ζ, ζ]



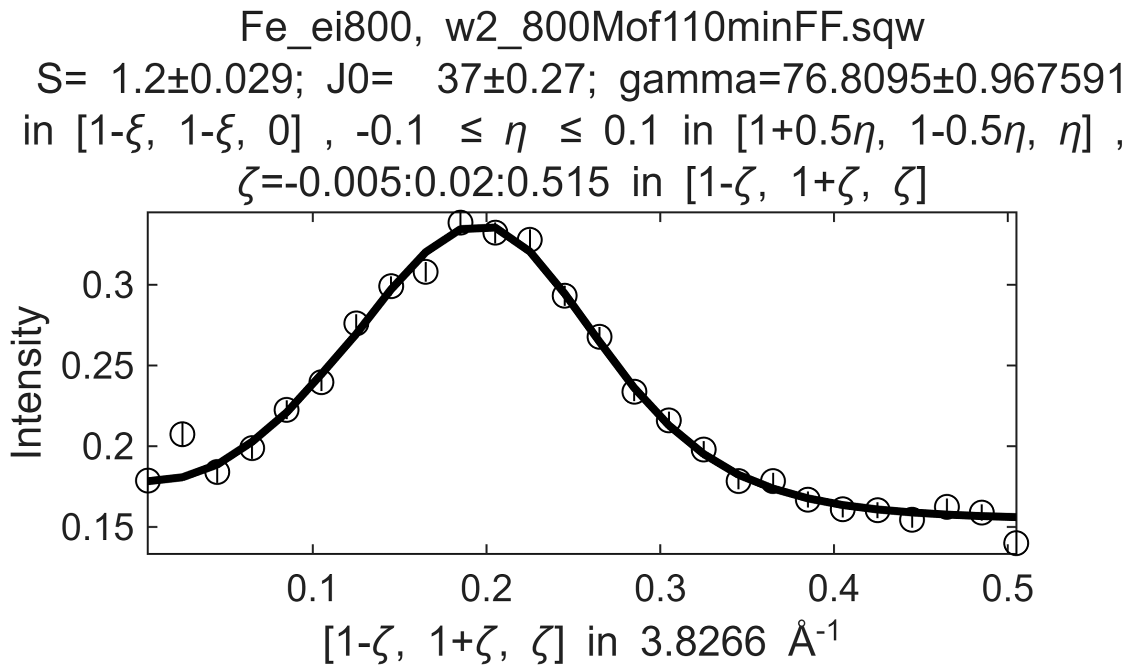


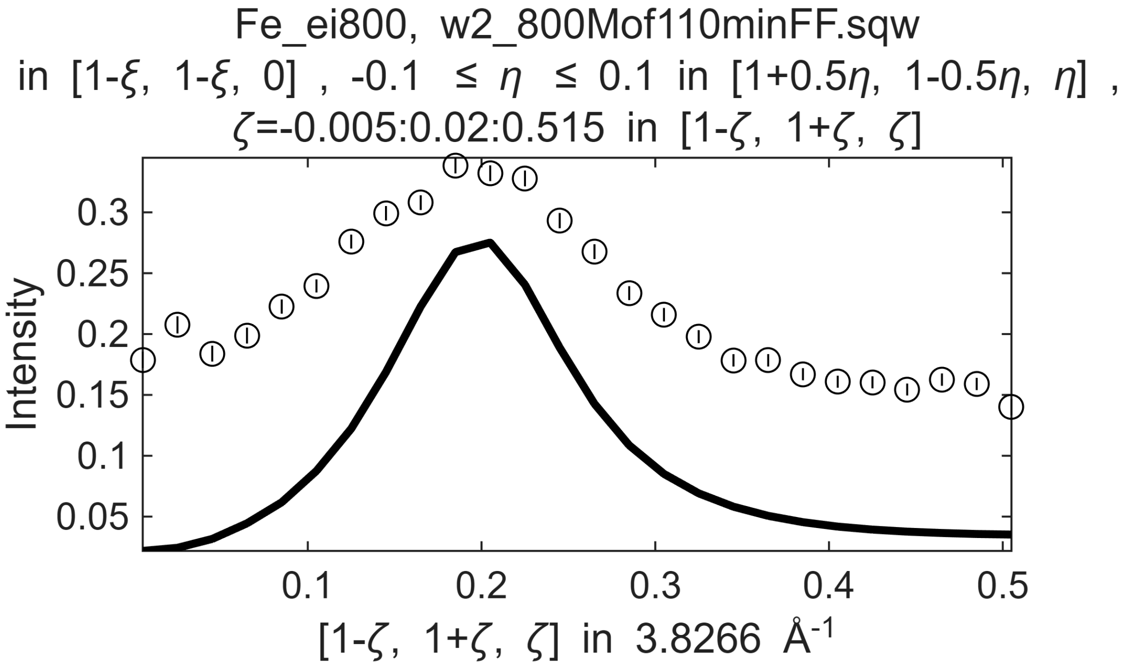

```
******************************
**** En = 125±5 going to 245
******************************
```



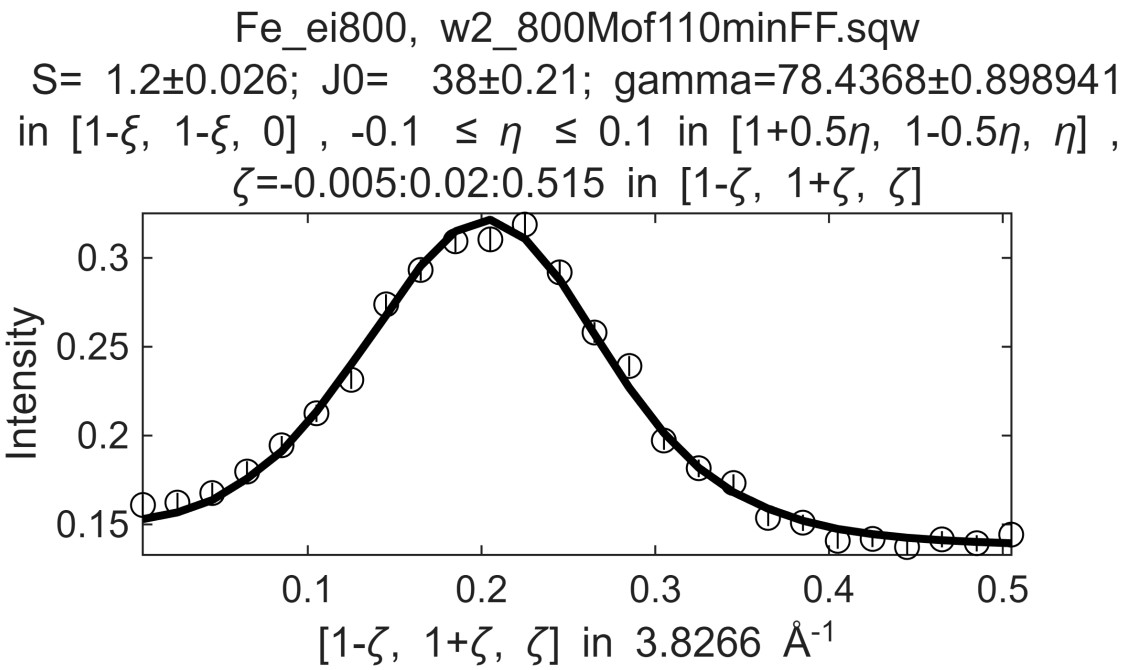



```
******************************
**** En = 135±5 going to 245
******************************
```

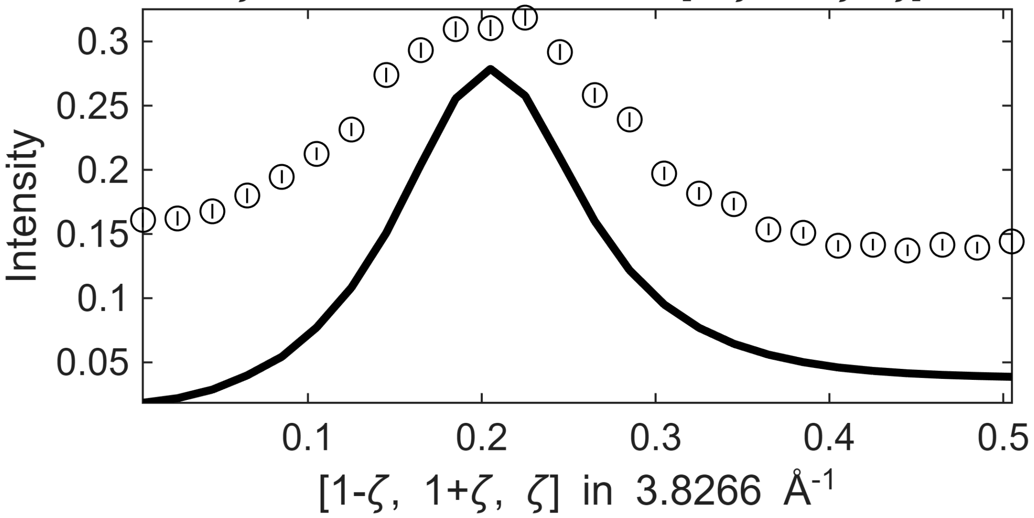




****
**** En = 145±5 going to 245
****

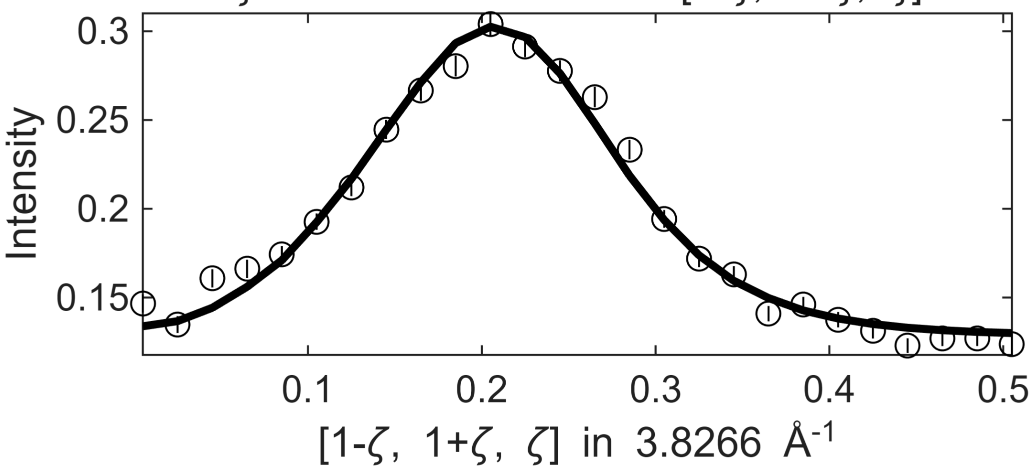


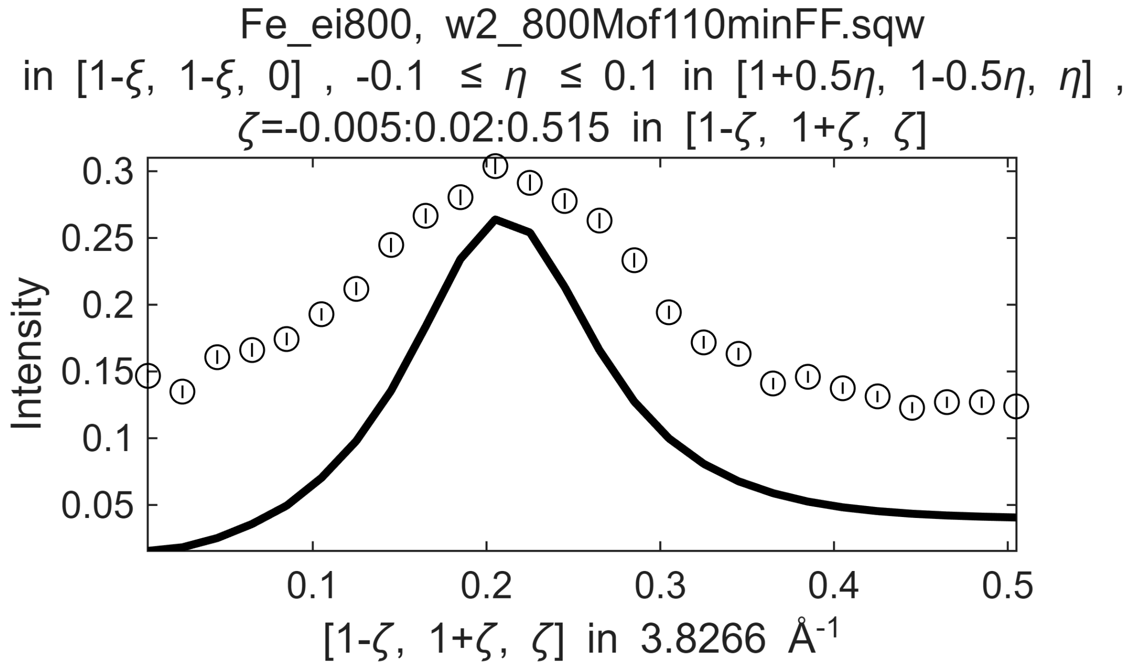
Fe_ei800, w2_800Mof110minFF.sqw
in [1-ξ, 1-ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1+0.5η, 1-0.5η, η] ,
ζ=-0.005:0.02:0.515 in [1-ζ, 1+ζ, ζ]



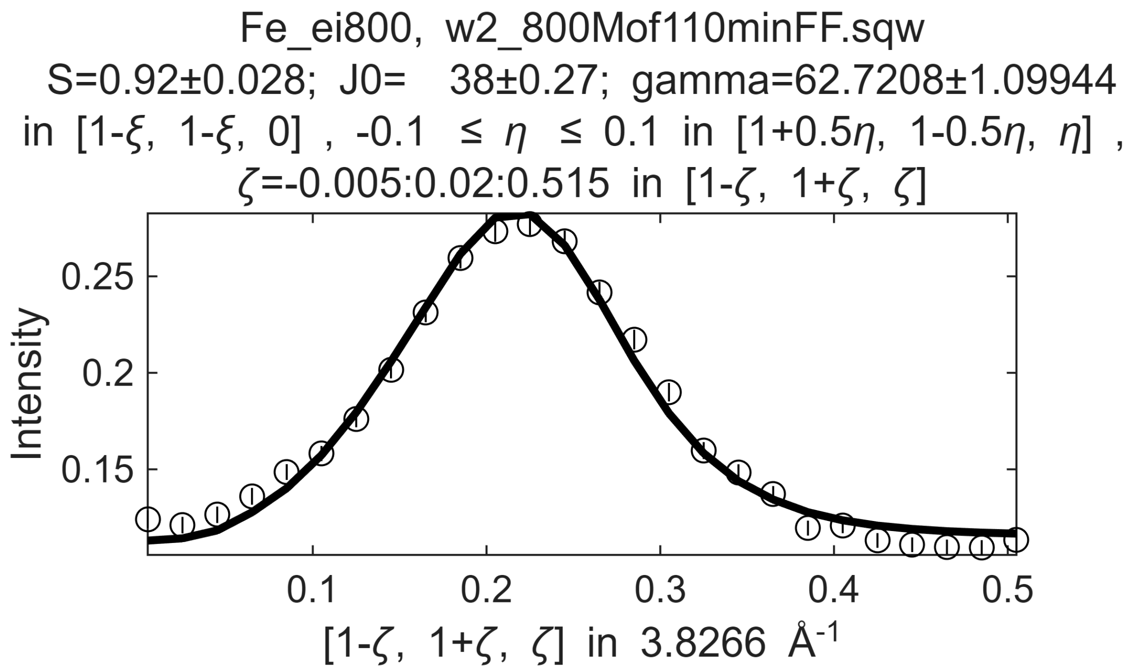

```
******************************
**** En = 155±5 going to 245
******************************
```

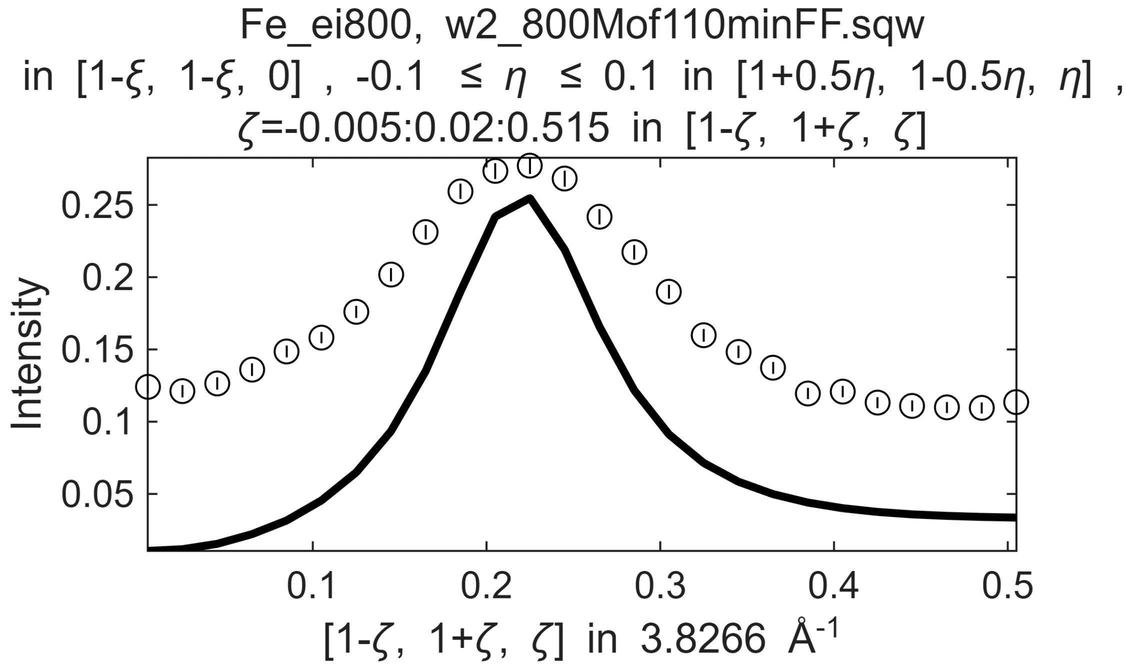
Fe_ei800, w2_800Mof110minFF.sqw
S= 1.2±0.04; J0= 39±0.28; gamma=81.2895±1.42807
in [1-ξ, 1-ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1+0.5η, 1-0.5η, η] ,
ζ=-0.005:0.02:0.515 in [1-ζ, 1+ζ, ζ]



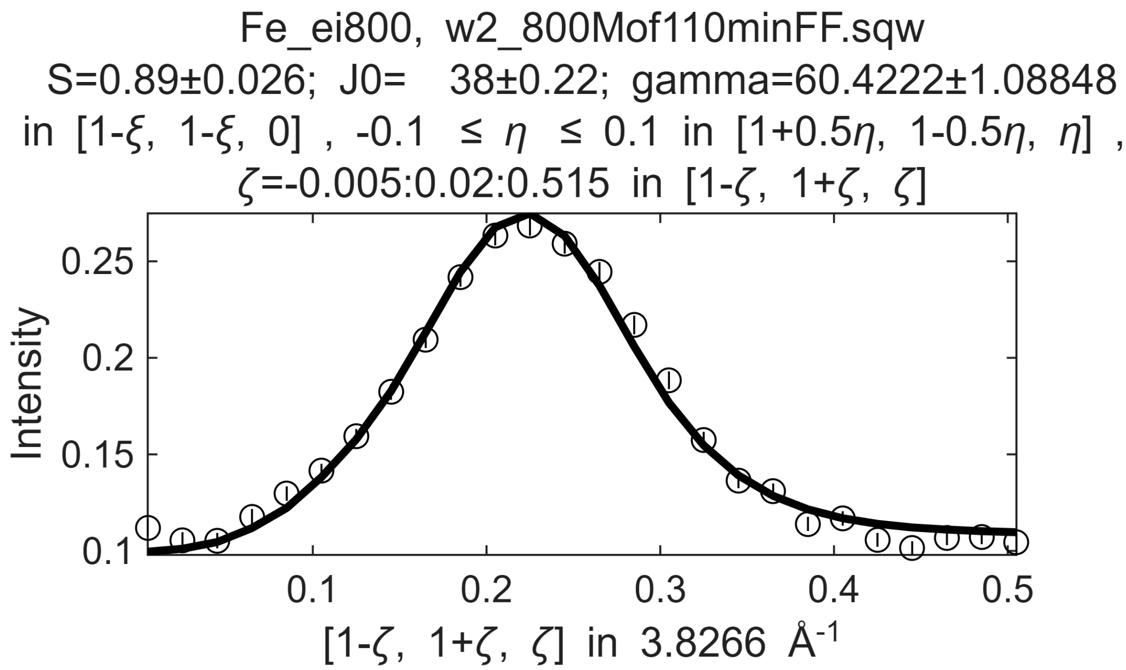


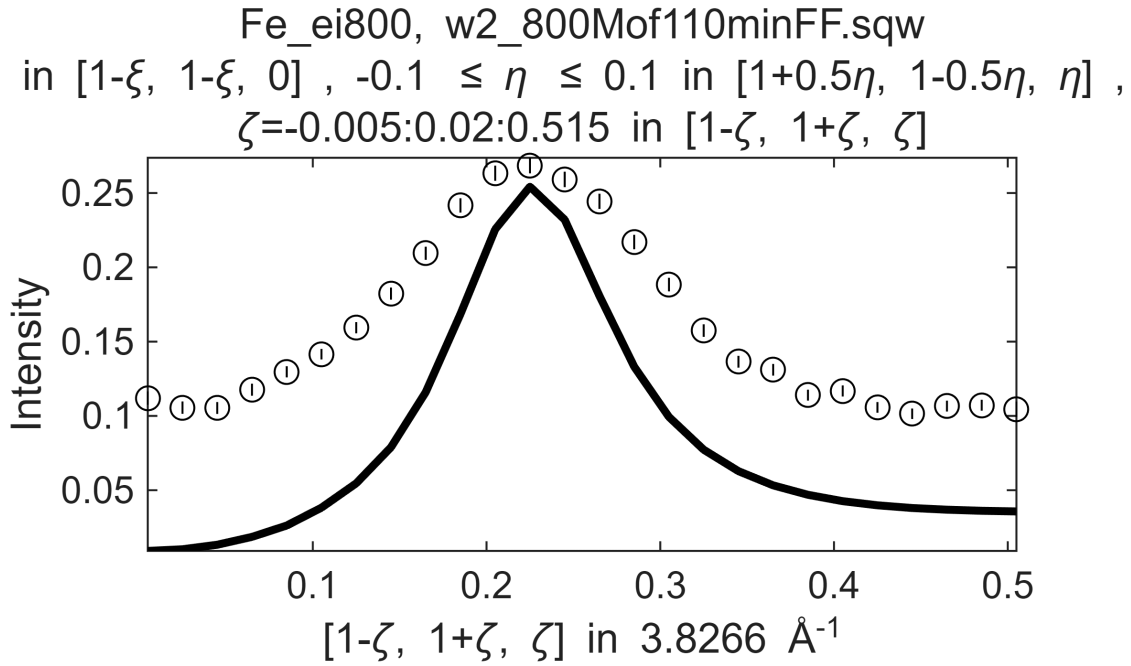
****
**** En = 165±5 going to 245
****



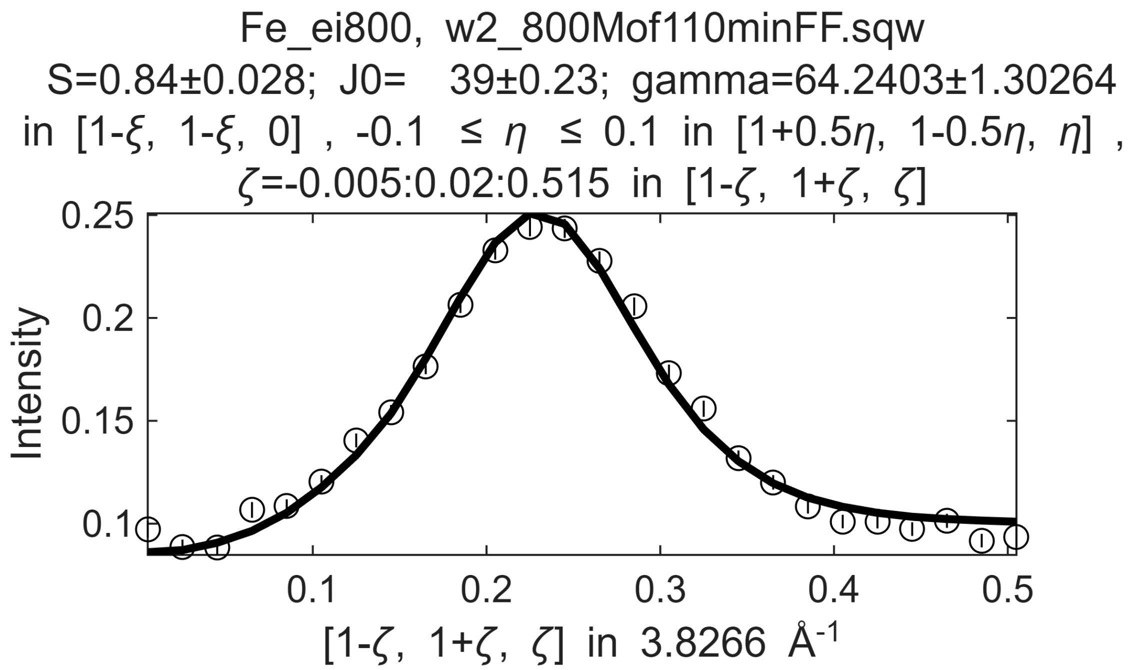


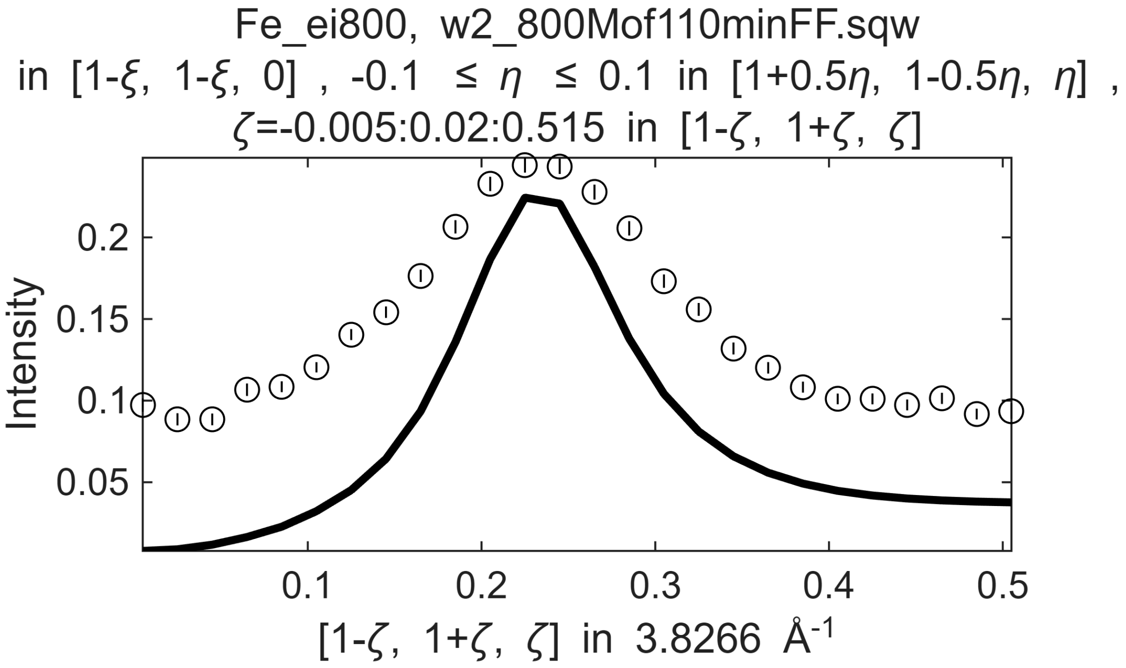

```
******************************
**** En = 175±5 going to 245
******************************
```



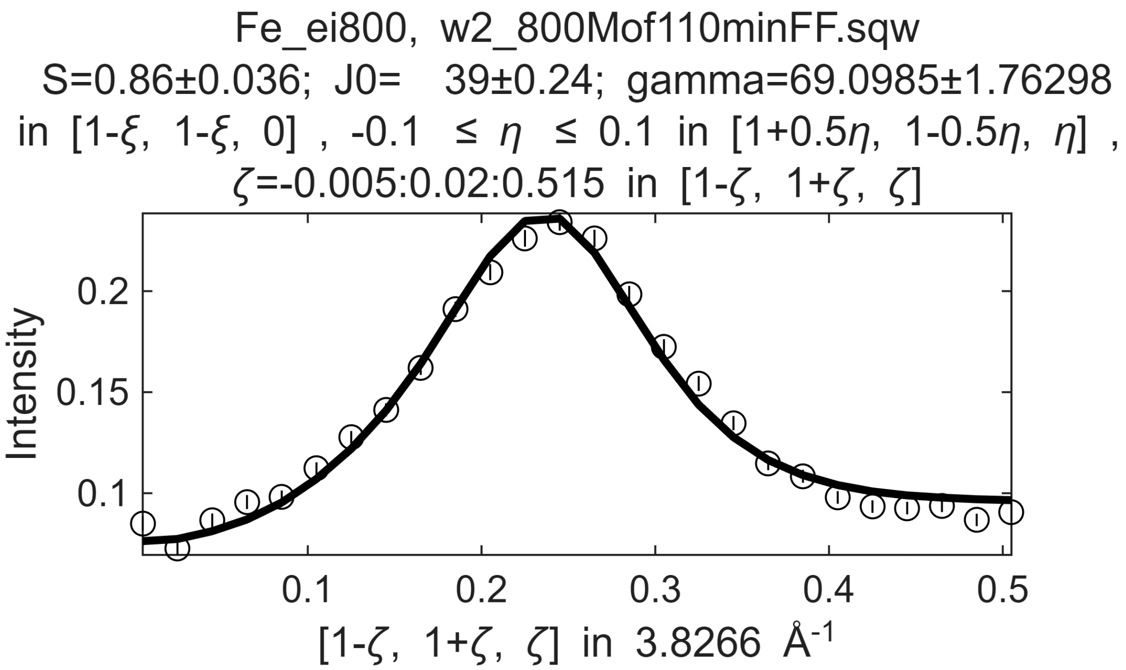



```
******************************
**** En = 185±5 going to 245
******************************
```

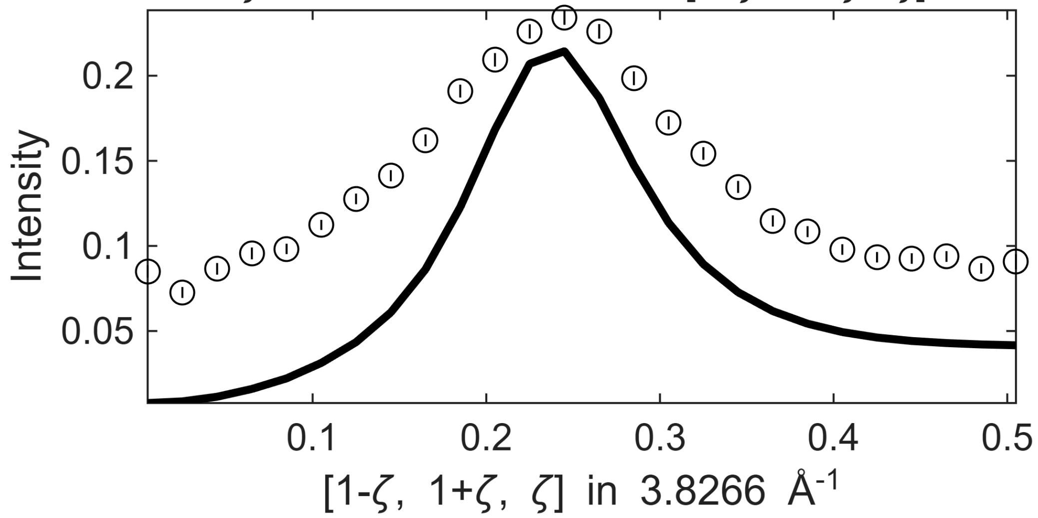





```
******************************
**** En = 195±5 going to 245
******************************
```

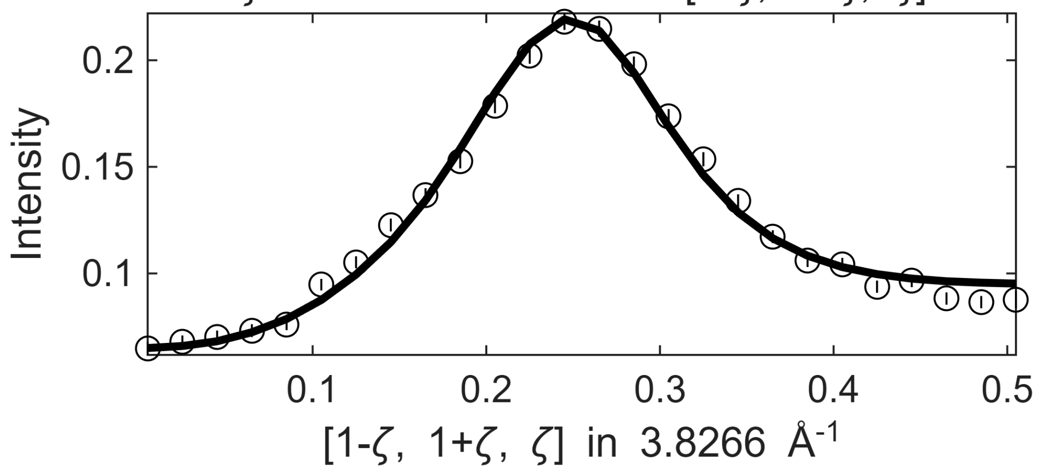


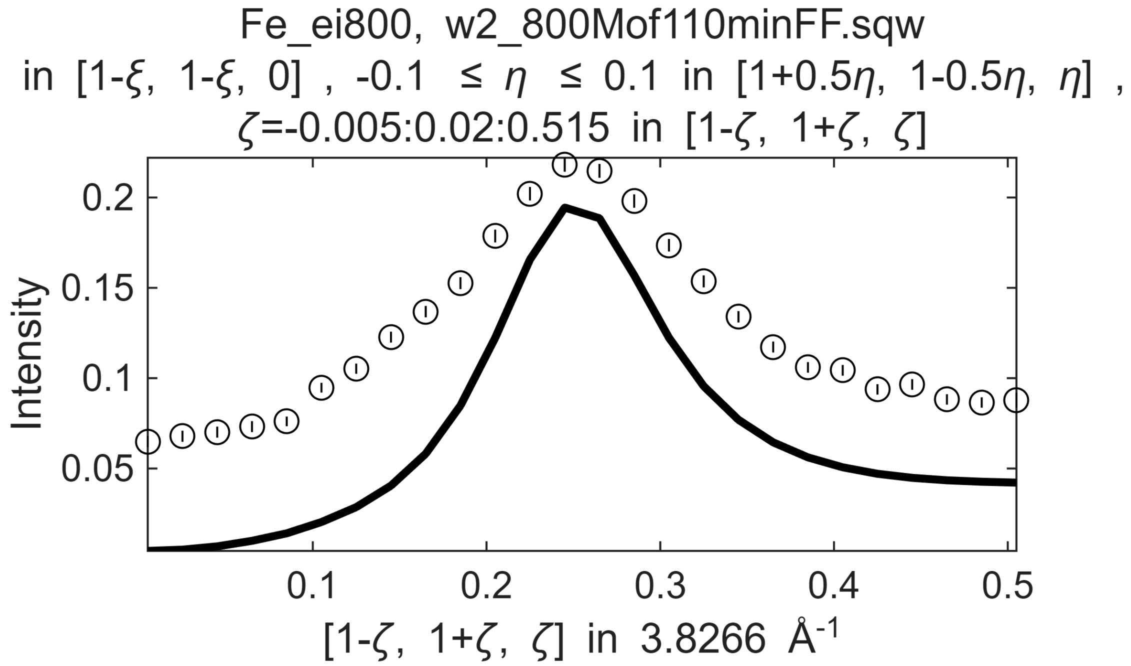
Fe_ei800, w2_800Mof110minFF.sqw
in [1-ξ, 1-ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1+0.5η, 1-0.5η, η] ,
ζ=-0.005:0.02:0.515 in [1-ζ, 1+ζ, ζ]



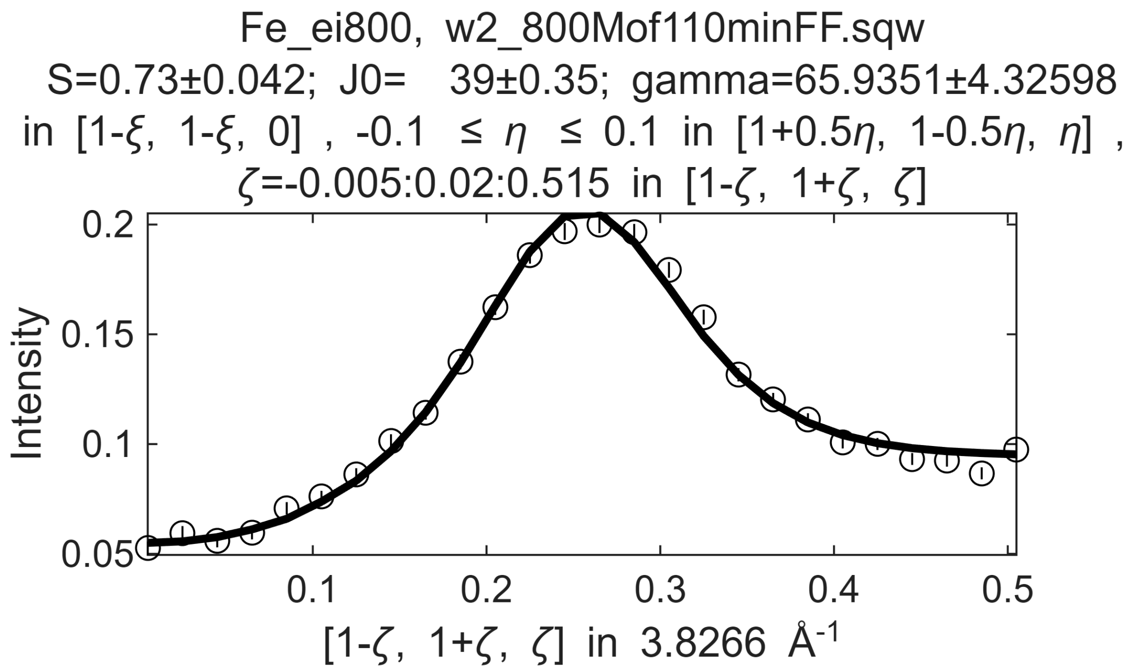

```
******************************
**** En = 205±5 going to 245
******************************
```

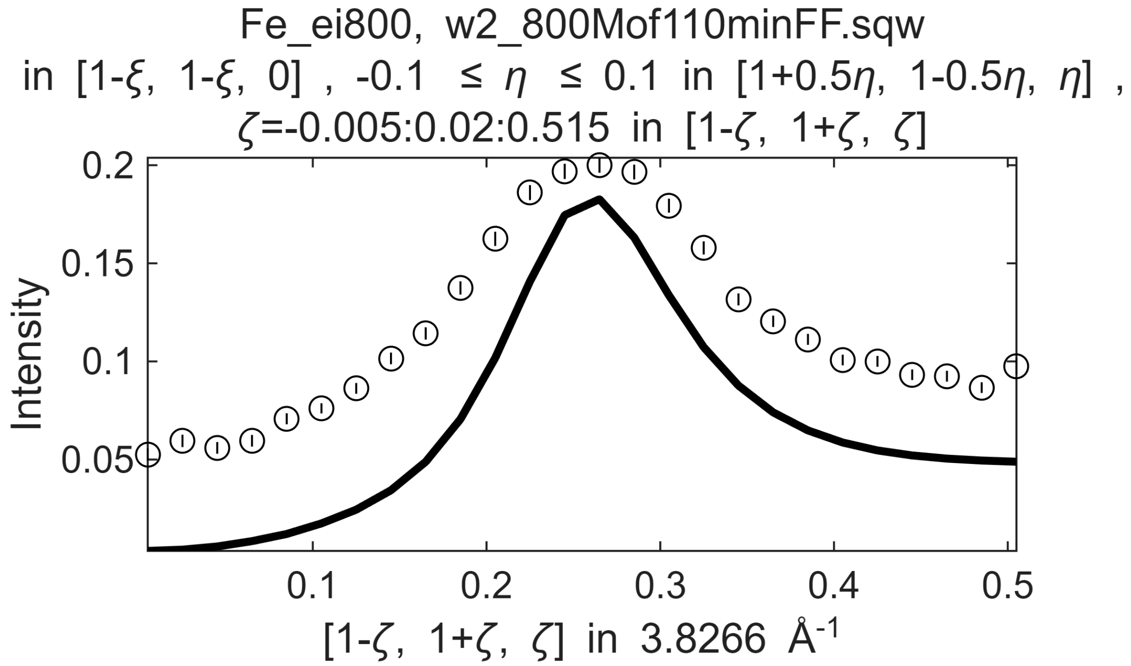
Fe_ei800, w2_800Mof110minFF.sqw
S=0.72±0.023; J0= 39±0.18; gamma=62.5271±1.26992
in [1-ξ, 1-ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1+0.5η, 1-0.5η, η] ,
ζ=-0.005:0.02:0.515 in [1-ζ, 1+ζ, ζ]



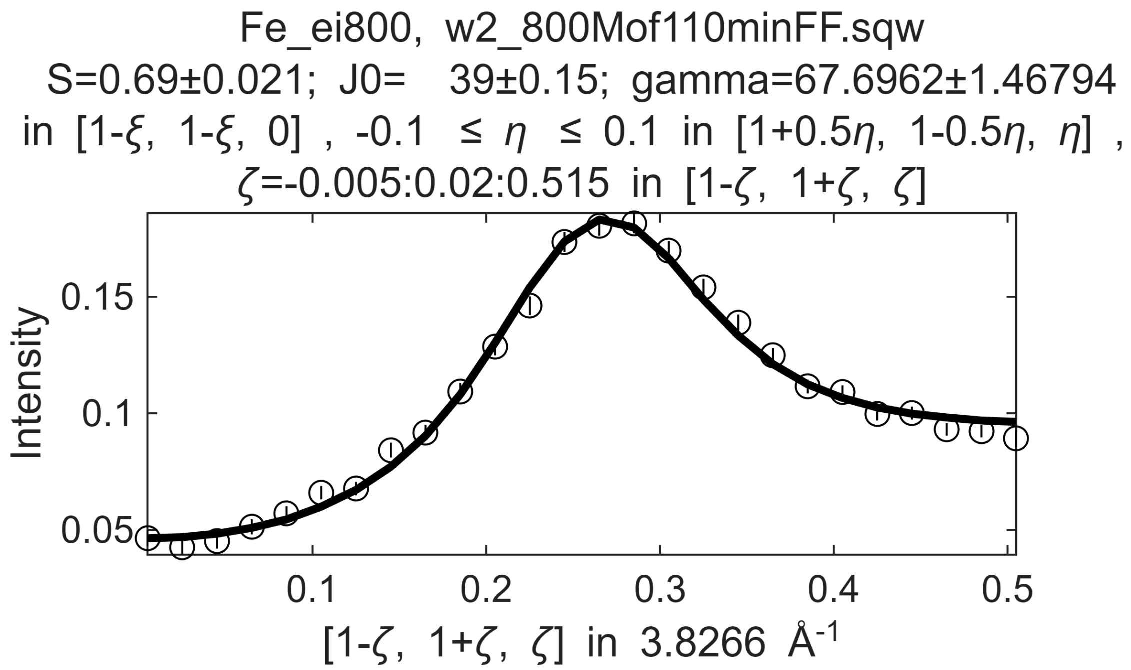


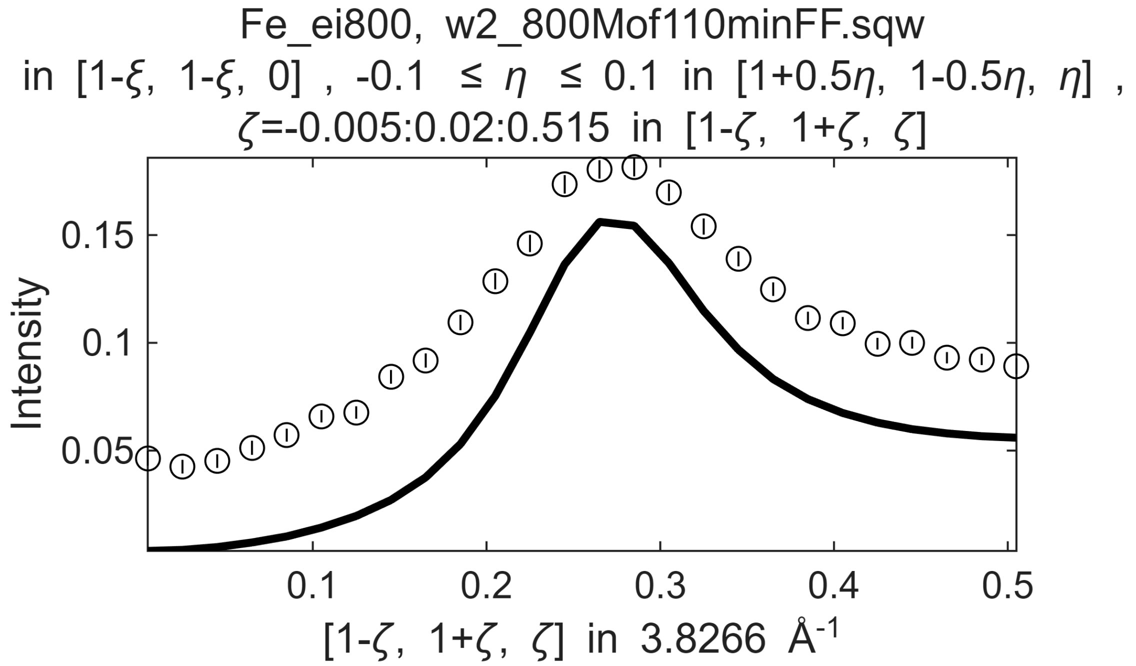

```
******************************
**** En = 215±5 going to 245
******************************
```



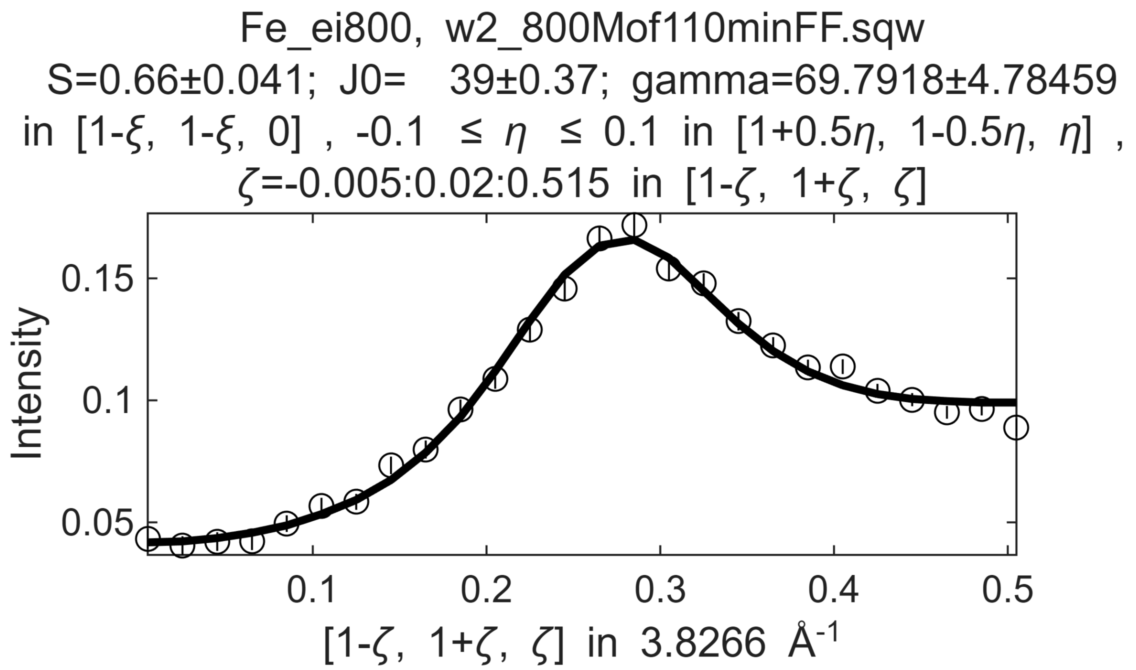



```
******************************
**** En = 225±5 going to 245
******************************
```

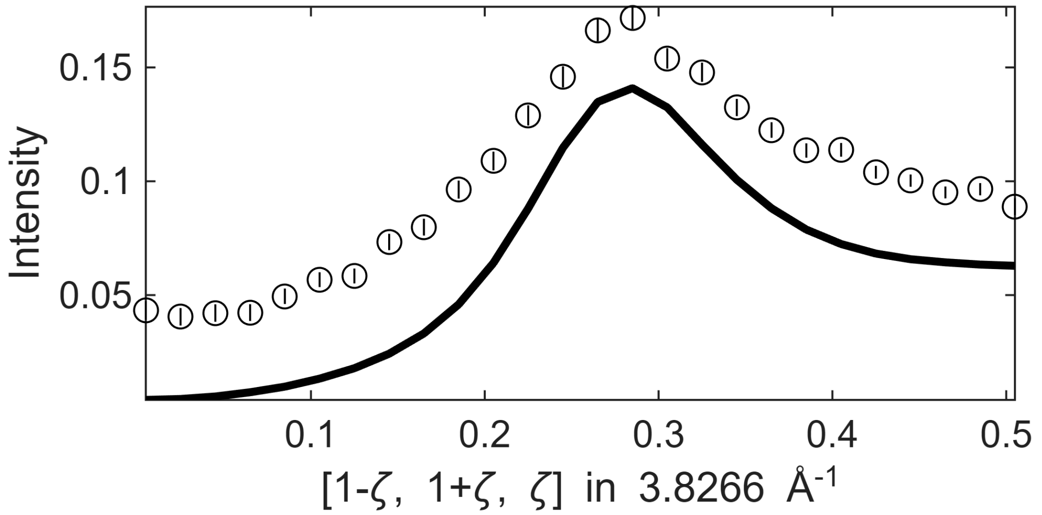





```
******************************
**** En = 235±5 going to 245
******************************
```

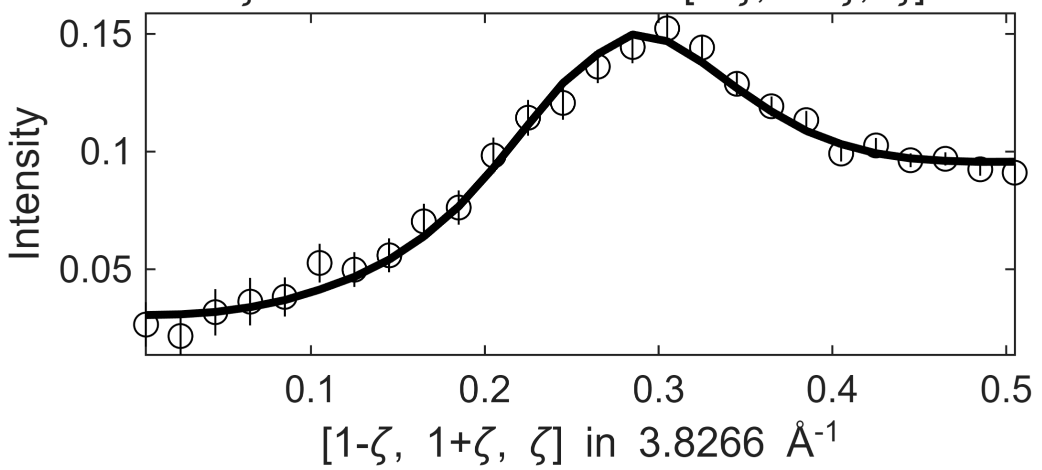


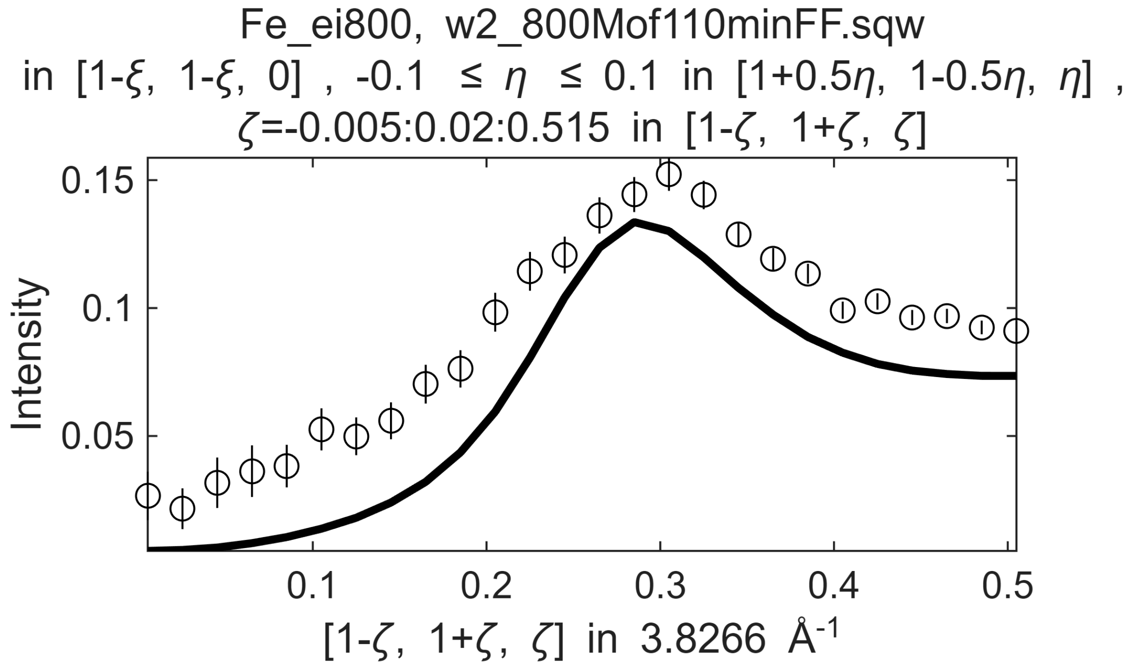
Fe_ei800, w2_800Mof110minFF.sqw
in [1-ξ, 1-ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1+0.5η, 1-0.5η, η] ,
ζ=-0.005:0.02:0.515 in [1-ζ, 1+ζ, ζ]

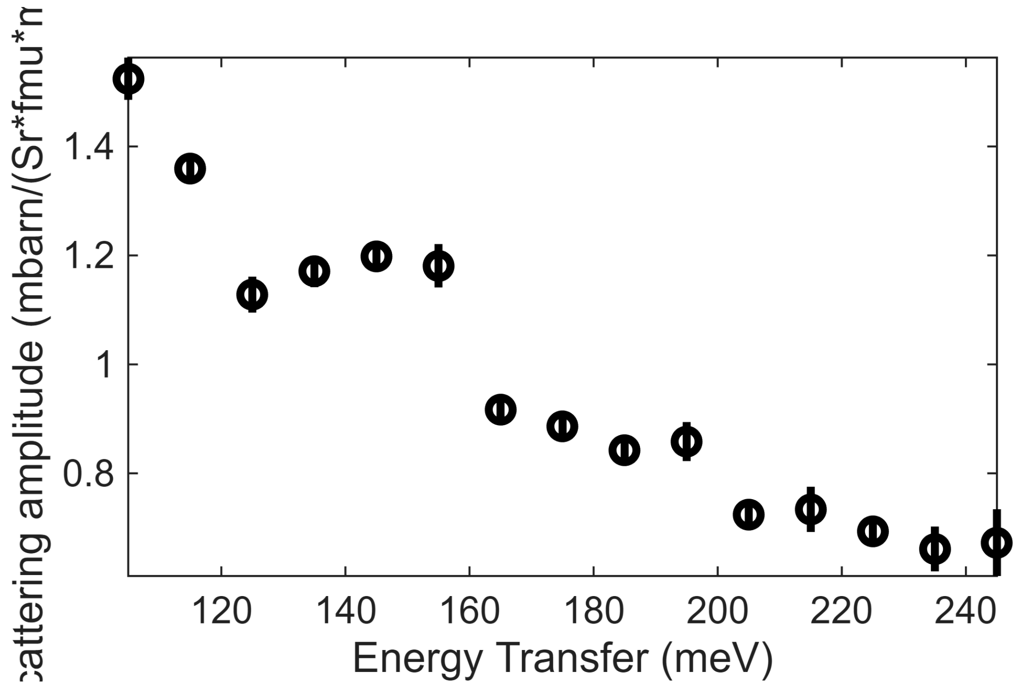


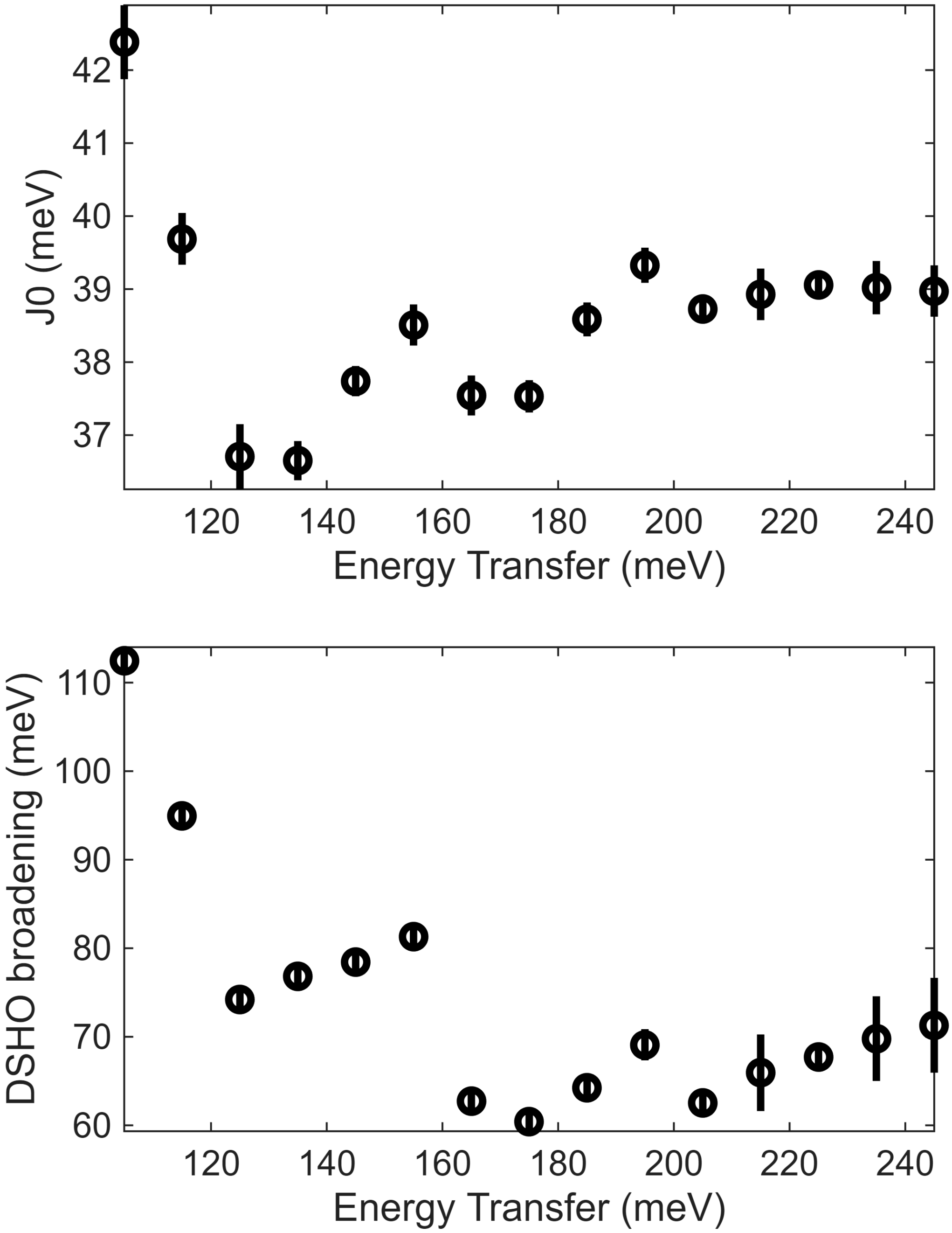

```
******************************
**** En = 245±5 going to 245
******************************
```

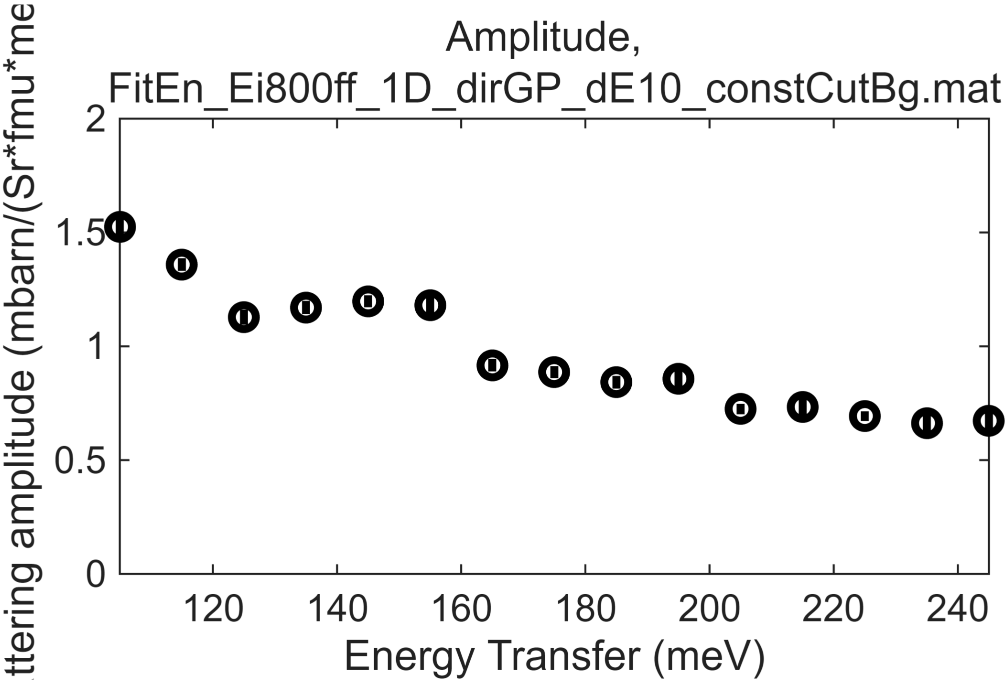
Fe_ei800, w2_800Mof110minFF.sqw
S=0.67±0.061; J0= 39±0.35; gamma=71.2991±5.3548
in [1-ξ, 1-ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1+0.5η, 1-0.5η, η] ,
ζ=-0.005:0.02:0.515 in [1-ζ, 1+ζ, ζ]

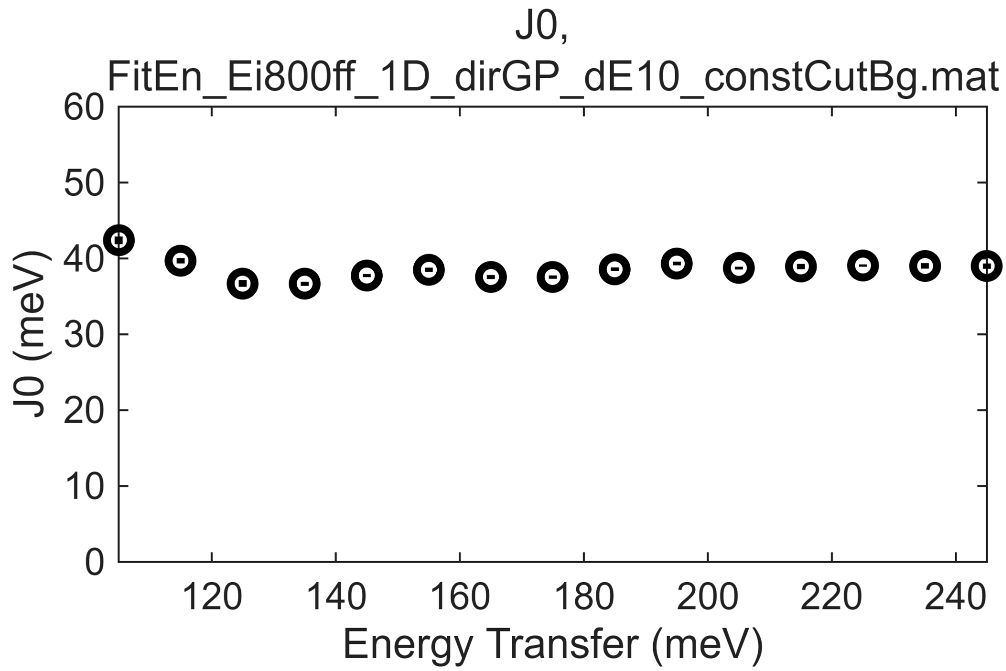


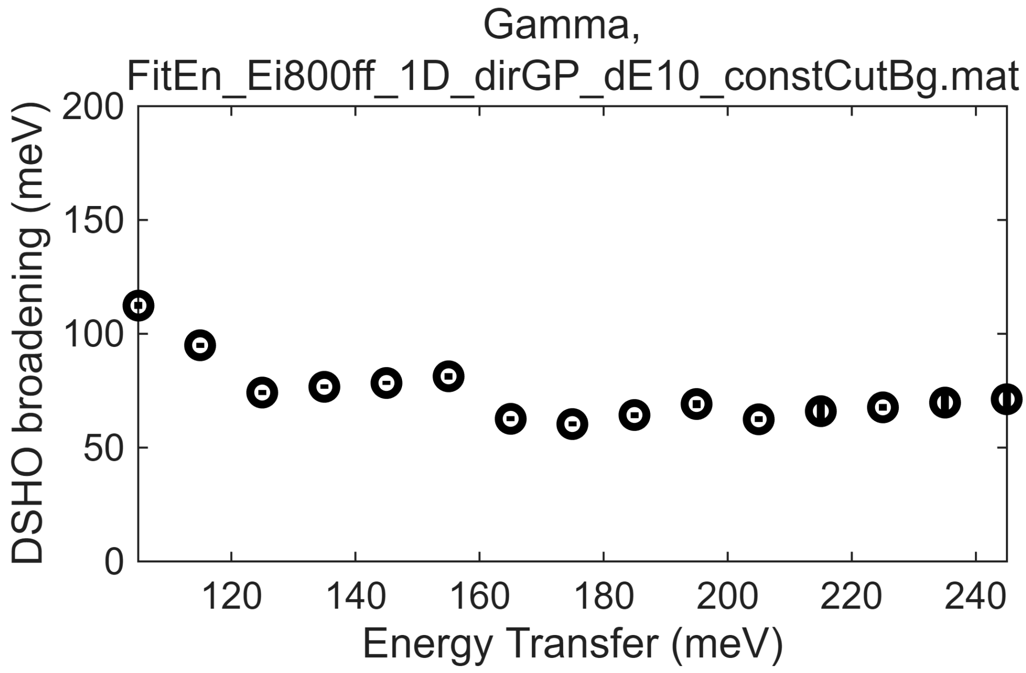

Fe_ei800, w2_800Mof110minFF.sqw
in [1-ξ, 1-ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1+0.5η, 1-0.5η, η] ,
ζ=-0.005:0.02:0.515 in [1-ζ, 1+ζ, ζ]
Intensity
0.15
0.1
0.05
0.1
0.2
0.3
0.4
0.5
[1-ζ, 1+ζ, ζ] in 3.8266 Å-1

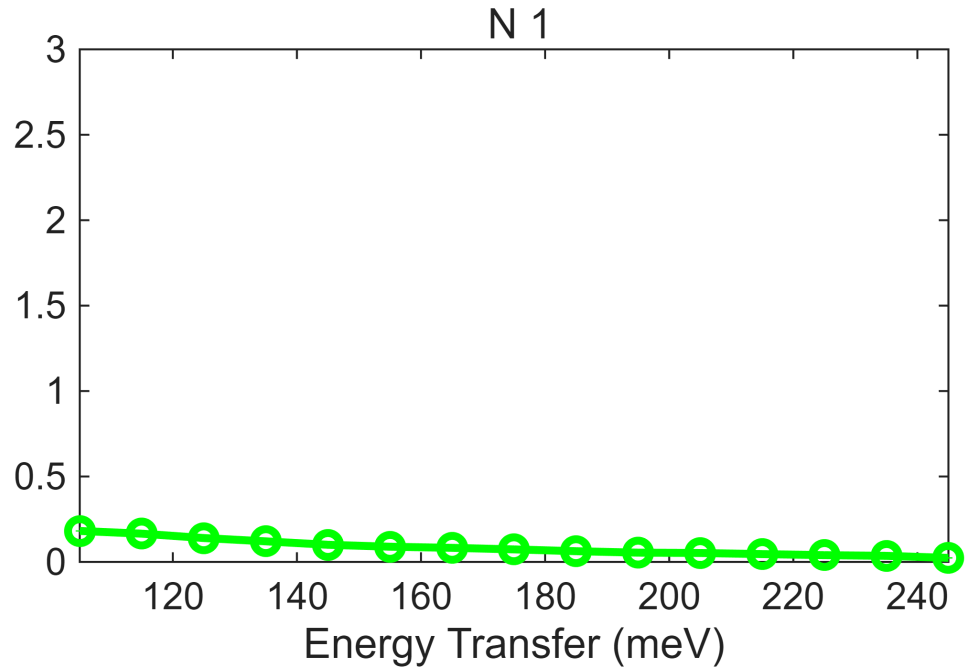

cattering amplitude (mbarn/(Sr*fmu*n
1.4
1.2
1
0.8
120
140
160
180
200
220
240
Energy Transfer (meV)

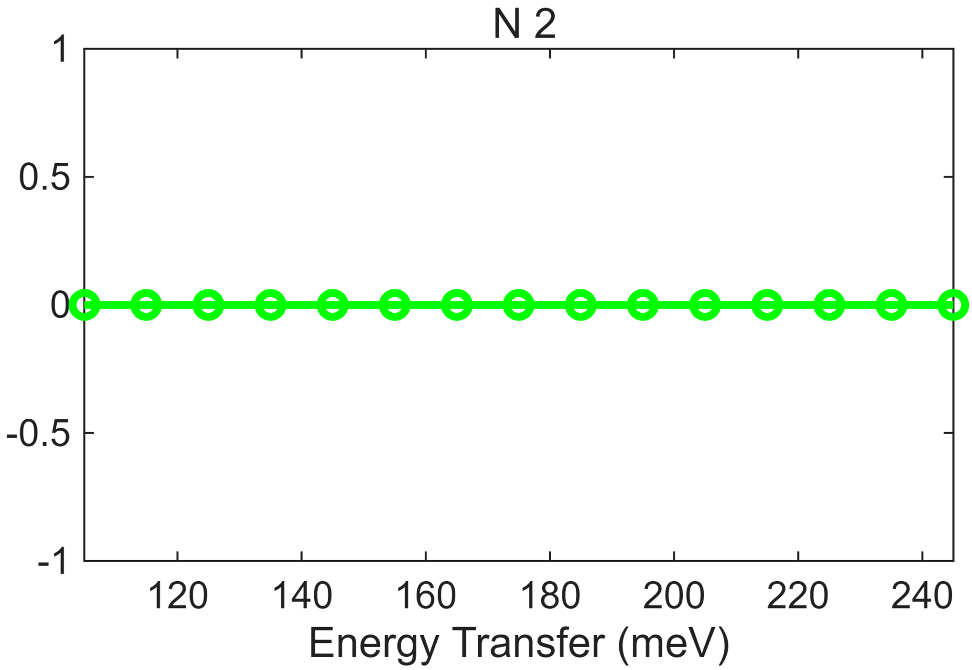

J0 (meV)
42
41
40
39
38
37
120
140
160
180
200
220
240
Energy Transfer (meV)
DSHO broadening (meV)
110
100
90
80
70
60
120
140
160
180
200
220
240
Energy Transfer (meV)


Amplitude,
FitEn_Ei800ff_1D_dirGP_dE10_constCutBg.mat
ttering amplitude (mbarn/(Sr*fmu*me
2
1.5
1
0.5
0
120
140
160
180
200
220
240
Energy Transfer (meV)


J0,
FitEn_Ei800ff_1D_dirGP_dE10_constCutBg.mat
J0 (meV)
60
50
40
30
20
10
0
120
140
160
180
200
220
240
Energy Transfer (meV)


Gamma,
FitEn_Ei800ff_1D_dirGP_dE10_constCutBg.mat
DSHO broadening (meV)
0
50
100
150
200
120
140
160
180
200
220
240
Energy Transfer (meV)


N 1
0
0.5
1
1.5
2
2.5
3
120
140
160
180
200
220
240
Energy Transfer (meV)




```
******************************
**** CUT: Ei800_off200maxFFGP
******************************
Error using plot
Not enough input arguments.

Error in plot_j0_fit_result (line 10)
plot(fit_res.S); ly 0 2;keep_figure;
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
Error in fit_multicuts_along_direction (line 23)
    fit_res = plot_j0_fit_result(ld.(fnms{1}),ref_name);
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```